



Intelligent mobile vending services: location optimisation for food trucks using coalitional game theory

Umang Kalra¹ · Ashish Kumar¹ · Tanmoy Hazra¹

Received: 11 August 2021 / Revised: 14 June 2022 / Accepted: 1 September 2022

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2022

Abstract

The food truck industry has seen substantial expansion in the past few years. Mobile service flexibility, low capital expenditure and overhead costs are some of the reasons for making it a popular venture choice to enter the hospitality industry. It is important for these mobile vendors to discover locations with high demand and purchase capacity, while facing low competition from competing eateries. A novel spatio-temporal multi-objective optimization approach, developed using a metaheuristic algorithm based on the principles of coalitional game theory, has been presented in this paper for allocating optimal retail locations in a spatio-temporal competitive multi-agent environment. Multi-agent competition has been modeled using a number of socio-economic parameters to better emulate the dynamic spatio-temporal behavior of urban crowds and buying power. The proposed algorithm has shown to solve the problem more effectively than existing combinatorial optimization algorithms. Moreover, the proposed approach shows how coalitional game theory can produce a set of Pareto optimal solutions for multi-objective optimization problems involving competitive multi-agent.

Keywords Combinatorial optimisation · Game theory · Food trucks · Multi-agent system · Metaheuristic algorithm

✉ Tanmoy Hazra
tanmoyhazra316@gmail.com

Umang Kalra
umangkalra10@gmail.com

Ashish Kumar
ashishkr0401@gmail.com

¹ Department of Computer Science and Engineering, Indian Institute of Information Technology Pune, Pune, Maharashtra 411048, India

1 Introduction

Ever since the land prices have started increasing exponentially and its unavailability at favorable places along with the time, effort and cost invested for the materialized building of a restaurant or a fast food shop, mobile food vending models have become a popular part of the hospitality industry. A number of studies show that the percentage would become higher if the food trucks were operated with smart, data-driven decisions [2, 6, 8]. Running a food truck business has advantages over traditional restaurant businesses, as food trucks can cover more demand if they follow population flow well. The fixed stores can only cover the population without movement, but mobile vendors can cover maximum population by following the population flows. Another major advantage is that people with the least investment can go into this business model, since it has relatively low cost for purchasing the food truck, as studied by Linnekin, Dermer and Geller [1]. However, to ensure profitability in running a mobile food vending business, a business framework has to be adopted, which takes important attributes like location and crowd trends. Factors like retail location, business ratings and competition from neighboring restaurants have a significant effect on the success of increasing sales, as found out by a number of studies [4, 25]. Based on this observation, Chang and Li [25] documented that user rating for a business is determined by both the intrinsic characteristics of the business like retail location, user ratings, food menu, demographics and the extrinsic characteristics of its geographical neighbors.

An intensive study has been carried out by Cho [14], where they modeled the problem of finding the optimal locations and routings for mobile vendors operating in Seoul, Korea. Similar research work has been done on food trucks operating in different cities, where this problem has been modeled as a *multi-agent spatio-temporal multi-objective optimization* problem [10, 15]. Factors such as location, neighboring competing restaurants and user ratings have been used to model the competition and object functions, so that optimal retail locations can be allocated dynamically. However, the existing research work for this problem still lacks inclusion of variables based on distance to be traveled by mobile vendors to reach the optimal location and dynamic competition from other mobile vendors operating in the same region. Our optimization approach overcomes this by using a multi-objective function which includes a competitive model based on neighboring restaurants, their ratings and competing food trucks, which is further discussed in section 3.

We have worked on a model which uses a heuristic algorithm derived from Coalitional Game Theory concepts. Studies reveal that optimization approaches based on *coalitional game theory* to solve the category of problem in discussion have shown satisfactory results [20, 38, 40, 47]. Burguillo and Juan define *game theory* as a study that provides a mathematical framework to formulate and analyze cooperative and competitive scenarios in an environment of several agents called *players* [41]. Research work has been conducted to show how various game theory models can be used either to create optimization algorithms or complement existing optimization algorithms to solve combinatorial optimization problems [5, 11]. Our proposed work is cornered with finding the optimal retail locations for food trucks or mobile vending services in an area, given that the real time data about buying demand of different areas of the city, competing food trucks and restaurant locations is provided. We have derived a cooperative-competitive model, which uses concepts of coalitional game theory, among the participating food trucks that uses factors like neighboring restaurants and food trucks, consumer demand at a location and distance a food truck has to travel to reach the location. A metaheuristic algorithm has been devised that uses this mathematical model to allocate

optimal retail locations to the food trucks. This algorithm has been then compared against popular optimization algorithms to assess the quality of derived Pareto optimal solutions.

2 Literature survey

Studies have shown that *location* plays a strategic role in the success for a restaurant business to stay competitive in the hospitality industry [17, 25, 44, 48]. Chua et al. investigated how the location of a restaurant affects choice and convenience of customers, and improves the visibility of a restaurant among its competitors [48]. Wang et al. studied the importance of *geographical placement* and devised a regression model that uses review-based market attractiveness and competitiveness, and geographic features of a location to predict the number of check-ins the restaurant can receive situated at that location [44].

The importance of a location in determining the food truck choice is as much for determining the response of customers towards choosing a restaurant. Choi et al. investigated the effect of selection attributes by different customer segments for predicting the dining intention towards food trucks [19, 24]. The selected attributes consisted of *location setting*, food and service quality, dietary considerations, previous customers' experience and food truck ratings. A large number of studies have been performed to understand the factors that drive customers towards choosing a food truck over restaurants [1, 16, 24, 30, 43]. However, very limited studies are available that explain the process of selecting an optimal retail location of a food truck. Cho developed a multi-objective spatio-temporal optimization approach that uses population distribution and predicted customer demand, and used mixed-integer programming to derive optimal locations [14]. However, their optimization does not consider the combined competition from nearby restaurants and food trucks, and the potential effect on consumer attraction that can arise from the neighboring food trucks. Similarly, Bhandawat formulated the problem of location optimization of food trucks to maximize their profits using a mixed integer programming model [15]. Nevertheless, it is not suited for real-time applications as it is very computationally intensive. Moreover, it does not consider the distance to be covered by a food truck to reach the optimal location. A balance has to be struck between consumer demand of the location and its distance from the initial position of the food truck. Cho performed an intensive study to formulate an optimization methodology to improve the spatio-temporal optimal location of a food truck by analyzing the geographic characteristics of the competing restaurants and the relationship between the de facto population [10]. However, the devised optimization methodology does not consider the *aggregation effect* that arises from multiple food trucks operating in the same location or very close to one another.

In order to add the effect of competition from competing restaurants and food trucks, a cooperative-competitive model has to be formulated that can be used for optimizing the retail locations. Wen et al. successfully modeled cooperative-competitive models for a multi-agent environment after investigating the competitive and cooperative trade-off among the participating agents [33]. A cooperative-competitive mathematical model is needed in place that guarantees not only a single food truck's profit, but of the entire multi-agent environment, that consists of all the food trucks operating in a particular region.

Multi-objective spatio-temporal optimization problems have been widely studied for a variety of scenarios [3, 13, 27, 28, 46]. These problems involve the optimization of multiple objective functions at the same time. It could be the maximization or minimization of all objective functions simultaneously or a combination of both. Multi-objective optimization

problems that are spatio-temporal in nature consist of *independent variables* that change with spatial & temporal differences. Uchigaito et al. formulated a multi-objective spatio temporal algorithm for optimizing power management and transport in a distributed management system consisting of multiple small grids which are installed in a small area in a distributed manner [13]. Trunfio proposed a multi-objective approach based on a genetic algorithm that uses spatio-temporal data for optimization of cellular automaton models [28]. Since the consumer trends are subject to change with spatial and temporal variations, the problem of optimal retail allocation for food trucks can be formulated and solved by multi-objective spatio-temporal optimization.

The goal of any optimization algorithm is to pick the best solution set from the set of an existing solution space. Game theory is the field of mathematics that provides the framework for analysis that can be used to make optimal selections. Sohrabi and Agzomi have reviewed the *random nature of game theoretic analysis* and how optimization algorithms and game theory have been used together to solve various problems, including multi-objective optimization problems [5]. Meng et al. designed a multi-objective optimization approach that uses *cooperative game theory* and showed its advantage over non-cooperative game theory in achieving collective benefit of the agents [36]. Coalitional or cooperative game theory provides the mathematical tools to find group efficient solutions for problems that involve strategic actions. Shams and Luise provided a study of coalitional game theory and its applications in resolving various problems that occur in communications and networking scenarios and how it can achieve improved performance when compared to solutions derived from non-cooperative game theory [34].

3 Proposed model

The proposed model aims at maximizing the profits of food trucks/mobile vending services and at the same time reducing the competition among the competitors. The participating food trucks/mobile vending services are referred to as agents in this multi-agent optimisation problem. The agents are allocated retail locations determined by factors like *profitability index*, as suggested in section 3.1, number of competing agents operating on those locations in a specific period of time. Agents competing for optimal retail locations over a common region form *coalitions*, as suggested in section 3.2. Coalitions is a group of players that is responsible for decision-making and enforces cooperative behavior. From the set of all possible coalitions, a stable *coalition structure*, as suggested in section 3.3, is chosen such that any agent from the set of all agents belongs to only one coalition and it cannot improve its profitability by a certain amount by changing its present coalition. The proposed algorithm assigns optimal retail locations to agents for all possible coalitions and finds the stable coalition structure, assigning the final retail locations to the agents which lie in the Pareto optimal solution set.

The assumptions that have been used for mathematical modeling are:

- Studies on consumer choice patterns have shown that consumers are inconsistent in choosing the same product or service frequently [12, 42]. This phenomena is based on changing choice conditions which affect their preference. That is why the mathematical framework of our approach is based on deterministic choice.
- The Effects of parameters like cuisine type of food trucks, pricing of food and their individual ratings are not considered in this model.

3.1 Profitability index

The city under study is divided into square cells of equal sizes forming a grid structure as shown in Fig. 1. Each cell is assigned a *profitability index*, calculated by using various decision parameters such as demographic and socio-economic data, number of restaurants and their ratings at a particular period of time, availability of parking space for the agents and popularity of that area at a particular period of time.

The aim of profitability index for a given cell is to provide an indication as to where the agents can maximize their profits on retail. This profitability index reflects the potential to attract customers in a particular area, which has a direct influence on the profits earned.

Bhandawat formulated an *attraction function* that reflects the potential of a location to attract customers [15]. The profitability index used in the proposed model uses similar attributes to rank different locations based on their potential to attract more consumers and less competition from nearby restaurants.

Let R_{gi} is the rating of i -th restaurant in cell g and c_g is the *popularity index* of the cell g . Then popularity index of a cell g can be found out by the following equation:

$$cg = \sum_{i=1}^m R_{gi} * p_i \quad (1)$$

where, m is the number of restaurants in a cell and p_i is the normalized number of visits to i -th restaurant during a particular period of time.

Let $P_{g,l,a}$ be the population density at cell g with income l and age group a , $I_{l,a}$ be the normalized scale of income level l and age group a and c_g (Eq. 1) is the popularity index of cell g . Profitability Index, $PI(g)$ for a cell g is calculated as:

$$PI(g) = cg * (\sum \sum P_{g,l,a} * I_{l,a}) / \sum_{i=1}^m R_{gi} \quad (2)$$

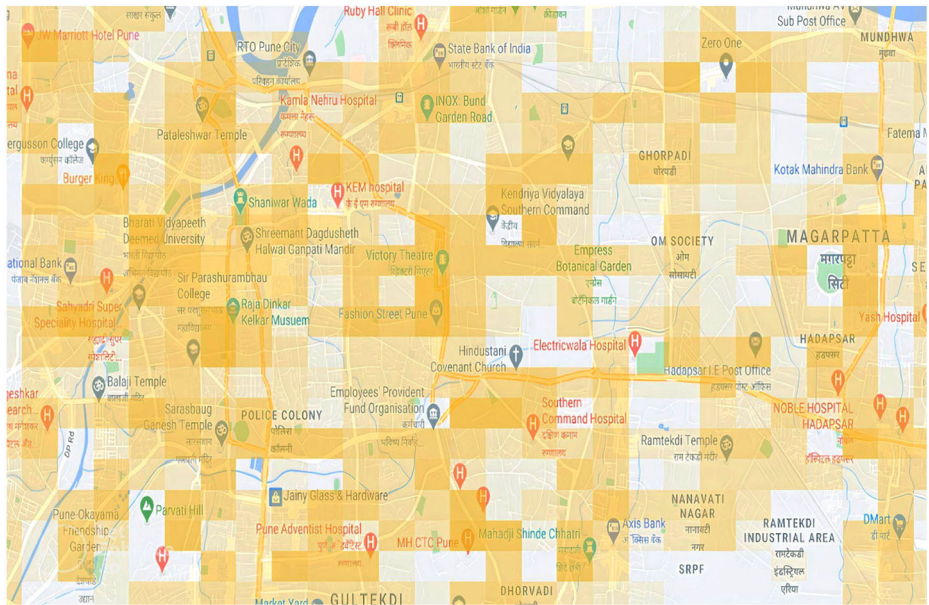


Fig. 1 The city under study is divided in grid structure, with each cell having its profitability index, PI . Darker shades representing higher PI and vice-versa

3.2 Coalition formation and location allocation

Let there be N food trucks. This optimal location allocation problem is a *multi-agent, combinatorial optimization* problem where the optimal set of locations is to be chosen for the participating food trucks. Solution to this problem can be derived by using a game-theoretic model. The food trucks are referred to as *rational agents*. Each agent has *preferences* and aims for the optimal location for increased profitability. Each agent has a *range* of operations i.e., it can only move to those locations that lie within its range. For each agent, there is a set of *actions*. Actions here refer to choosing a coalition to join and follow the cooperative contract made between the coalition and accept the allotted location for retail by the algorithm. Each agent also has a *payoff function* that maps the agent's payoff to every action that is available to that agent. These payoff values do not indicate the actual profits that are earned by the agent on taking a particular action, but rather it is a quantity to signify the preference among different available actions.

Let A_i be the set of actions available for agent i , where $i \in N$. A_i contains all the locations that agent i can choose for retail. Let u_i be the payoff function for agent i . The preference of agent i to an action a over b , where $a, b \in A_i$, is represented by payoff function. If $u_i(a) > u_i(b)$, then action a is preferred over action b by agent i , or location a is given more preference than location b .

An agent's strategy is defined as the action they choose in a setting where the outcome depends on their actions as well as on the actions of other agents participating in a game. To find the optimal retail location, at a given period of time, for a mobile vending service, optimal strategy has to be chosen.

Nash equilibrium is a theorem used in game theory that defines that a player can achieve their target outcome by not changing their initial strategy [35]. For our model, *Cooperative Game Theory* has been applied. The motivation behind using this model is the fact that in many games, Nash Equilibrium yields non-optimal outcomes compared to non-equilibrium outcomes [23, 37]. In cooperative game theory, agents form coalitions with the goal of securing better payoffs. The agents within a coalition can communicate with each other, formulate strategies following a *contract* or a *correlated strategy*. Agents that sign a *contract* are required to take their decisions according to an identified strategy which is known as *correlated strategy*.

The definition of payoff function u_i for an agent i , using the assumptions mentioned in section 3, is given as:

$$u_i(a_i) = (PI(a_i)/n) + r - d_i \quad (3)$$

where, $PI(a_i)$ is the *profitability index* of location a_i , $a_i \in A_i$, n is the number of agents that are assigned the same location as agent i , r is the agent's range of operation, d_i is the distance between a_i and agent's starting position, given by attribute of agent *init_location*. d_i is calculated as:

$$d_i = \|a_i - i.init_location\| \quad (4)$$

where, a_i and $i.init_location$ are agent i 's assigned location and starting position respectively. d_i is the norm of a_i and $i.init_position$. Locations are represented as 2-D vectors and d_i represents the Euclidean distance.

The following *contract* is followed by every agent that takes part in a coalition C :

- If the coalition is singleton, that is, it contains only one agent. In this case, the *correlated strategy* is a_i such that $u_i(a_i) > u_i(a_j)$, $a_i \neq a_j$, $\forall a_i, a_j \in Z_i$, where

$$Z_i = \left\{ A_i - \bigcup_{k \neq i}^N \left((A_i \cap A_k) \right) \right\} \quad (5)$$

If $Z_i = \{\emptyset\}$, then there is no correlated strategy. Then payoff would be set to a highly negative value, to show that it is the least desirable action, as shown below:

$$u_i(\{\emptyset\}) = -\infty. \quad (6)$$

- If the coalition is not singleton, then the *correlated strategy* for coalition C , $S(C)$ is given by the heuristic algorithm, whose pseudocode is given in [Algorithm I]. It stores the optimal retail location for an agent in its attribute *retail_location*, which is a dictionary with keys as coalitions and their corresponding values as 2D vectors representing locations. The payoff is also stored in an attribute called *profit*, which is also a dictionary with keys as coalitions and their corresponding values as payoffs that an agent earns by participating in a coalition. This algorithm sorts the agents in coalitions by increasing the order of their best potential payoffs over the action set X , with $n = 1$ in Eq. (3). Then, the agent is allocated the location which maximizes its payoff, while keeping a record of the locations assigned and the number of agents assigned to such locations. This process is repeated for all the agents in the coalition. X is given by:

$$X = \bigcap_i^C A_i - \left\{ \bigcup_j^{N-C_k} \bigcup_j^C (A_k \cap A_j) \right\} \quad (7)$$

There might also be a situation where some of the agents of the coalition might earn better payoff values by choosing the locations from their respective independent regions. In this situation, they allow other agents of the coalition to find better retail locations. This decreases the competition among them and increases profitability for all agents in the coalition. This algorithm sorts the agents in coalition.

Let $S(C)_i$ be the correlated strategy for agent i , where $i \in C$. If agent i opts out and decides to operate in its own independent region, the new $S(C)$ would be:

$$S(C) = (S(C) - S(C)_i) \cup S(\{i\})_i \quad (8)$$

Algorithm 1 Location Set Allocation to Coalition**Input:** A coalition of agents C ,*Initialisation:*

1. **for** agent in C **do**
2. $X \leftarrow X \cup A_{\text{agent}}$
3. $Z \leftarrow \{\}$
4. **for** remaining_agent in $(N - C)$ **do**
5. **for** agent in C **do**
6. $Z \leftarrow Z \cup (A_{\text{remaining_agent}} \cap A_{\text{agent}})$
7. $X \leftarrow X - Z$
8. Create a dictionary $Y \leftarrow \{x : 0, x \in X\}$, to count number of agents on the same location
9. **for** agent in C **do**
10. agent.init_payoff = $\max(u_{\text{agent}}(x))$, where $x \in X$, $n = Y[x] + 1$
11. sortedCoalition $\leftarrow C$ with agents sorted according to agent.init_payoff in non-decreasing order.
12. **for** agent in sortedCoalition **do**
13. **for** x in X **do**
14. $n = Y[x] + 1$
15. agent.retail_location[C] = $\arg \max_x (u_{\text{agent}}(x))$
16. $Y[x] = Y[x] + 1$
17. **for** agent in sortedCoalition **do**
18. $n = \text{agent.retail_location}[C]$
19. agent.profit[C] = $u_{\text{agent}}(\text{agent.retail_location}[C])$
20. **if** agent.profit[C] < agent.profit[{agent}] **then**
21. agent.profit[C] = agent.profit[{agent}], $Y[\text{agent.retail_location}[C]] -= 1$,
 agent.retail_location[C] = agent.retail_location[{agent}]

3.3 ϵ -Core of the game

Transferable utility is a widely used concept in cooperative game theory. Utility is said to be transferable if the utility or value of an agent can be transferred completely or a part of it to another player [21]. In terms of coalitions, all the members of the coalition enjoy the total utility, irrespective of the individual division of utility. This game in discussion is essentially an NTU(Non Transferable Utility) game because the demand attracted by the agents for retail is not transferable.

In cooperative game theory, a *core* is the set of all feasible allocations such that no agent can acquire a better payoff by deviating from it or by changing their coalition. To find the optimal retail location allocations, it is required to find the core of the game, but the problem of instability and computational inefficiency can arise for larger games. They can be compensated to some extent by using the concept of ϵ -core, which is an approximate solution and widely used to solve computationally large problems [31].

A *coalition structure* is a partitioning of a set of agents N into a set of disjoint coalitions. Different coalition structures may have different action sets. To find the ϵ -core, the concept of *Minmax Coalition Formation*, as described in [49], is used. Using this concept, a coalition structure and its corresponding action set can be found in the ϵ -core, where no agent chooses to *defect*. An agent i of coalition C with correlated strategy $S(C)_i$ can *defect* if it strictly prefers coalition D with correlated strategy $S(D)_i$, such that $u_i(S(D)_i) > u_i(S(C)_i)$. In ϵ -core, no agent

can earn a better payoff value by ε . The coalition structure with the minimum *Maximum Regret*, as suggested in [49], is the *minmax coalition structure* with corresponding regret as *minmax regret*. *Regret* is the loss of utility/payoff value that an agent faces by accepting a coalition structure and not defecting. This *minmax regret* is equal to ε . If $\varepsilon = 0$, then it would be the core of the game.

Let K be a coalition structure. The coalition to which agent i is a member, given by K_i and the corresponding correlated strategy for coalition K_i is given by $S(K_i)$.

Let the maximum payoff value that any agent can get is stored in its *best_payoff* attribute. Then maximum regret, MR of a coalition structure K is given by:

$$MR(K) = \max_{i \in N} (i.best_payoff - u_i(S(K_i))) \quad (9)$$

All possible coalition structures are formed from all the coalitions that have been made. For each such coalition structure, maximum regret is calculated using Eq. 9. The coalition structure with the minimum maximum regret, is declared as the ε -core of the game and the correlated strategies made according to this coalition structure are assigned to every agent, which is basically optimal retail locations for participating food trucks.

4 Experiment and results

To verify the proposed approach and test its efficiency, the designed experiment is based on a large region of Pune city, India. A map of this area was extracted from Google Maps and it is divided into equal sized cells forming a Fishnet structure. This grid spans over an approximate area of $10 \text{ Km} \times 7 \text{ Km}$. There are 29 cells in each row and 21 cells in each column. Therefore, each cell is roughly of the size $340 \text{ m} \times 260 \text{ m}$. Each cell has been assigned a profitability index, *PI* based on the parameters discussed in section 3.1. *QGIS* has been used to create the map [26]. The map of the city under the city was fetched using Google Maps API [39] and a raster layer was created with *QGIS* where each cell had been assigned a profitability index. A synthetic dataset has been created from the demographic, socio-economic data and the realtime information of the local restaurants for calculating the *profitability index* has been gathered from Google Trends data and the database of the City Census Department of Pune [29]. The data was gathered for the time period from 4:00 pm - 6:00 pm on 2nd October 2020. In a real-time application, periodic data would be fed to the system, to calculate updated profitability indices so that new optimal retail locations can be found. In this experiment, only the data of the mentioned time interval is used. The cells in the map are colored according to their profitability indices, darker shades indicating higher profitability index and vice-versa. Figure 1 shows the complete map of the area under study, with each cell colored to indicate the magnitude of *profitability index*.

The operating conditions of the food trucks are kept the same for all the different test cases. The range of service has been set to 3 km. Different test cases involve the change of starting positions of the food trucks only. The profitability index map remained the same. To compare the proposed algorithm, Genetic Algorithms [9] and Tabu Search [32] have been selected from a wide range of optimization algorithms. Both these algorithms have been used widely to solve various combinatorial optimization problems, given the fact that [5]. A genetic algorithm replicates the process of searching and optimisation by imitating natural biological processes

for instance mutation, selection, inheritance and crossover. The algorithm initializes a random population of *genomes*, which belong to the solution set of the problem. Then a sequence of new populations is created as it progresses. Individuals of the current population are used to create individuals of the next population by performing crossover between them and applying mutation to reduce the possibility of getting stuck in local optima. Permutation based encoding is used to perform permutation based crossover between the genomes [22], as *genes* represent locations in the area and they were made up of 2D vectors [45]. Scramble mutation is introduced to the individuals at each step. In this mutation, from the genome, a subset of genes is selected and their values are scrambled randomly.

Tabu search is a metaheuristic search method. It uses a guided local search method designed to avoid local optima and moving past already visited search space points with the aim of obtaining an optimal solution. Its performance is enhanced by introducing a special rule that forbids it from considering a potential solution if it has been previously visited within a certain short-term period. It performs by maintaining a list of such search points, called the *Tabu List* [7].

Renman and Fristedt studied the performance of genetic algorithm and tabu search and compared the acceptability of the generated solutions by comparing metrics like mean, standard deviation, number of iterations and the time taken to generate the optimal solutions [7]. The relationship between time taken and number of iterations varies depending on the computational expenses of the steps involved. Therefore, for our experiment, the following metrics have been considered for evaluation for the efficacy of the algorithms:

- Mean of the payoffs that the agents would get for choosing to relocate to the allocated location.
- Standard deviation of the final payoffs has been used as a metric to measure if the allocations are fair and not benefitting a few members in the coalition, including the complete multi-agent environment.
- Time taken by the system to generate the solution set for each algorithm considered under experimentation.

A higher mean would suggest that the food trucks are earning better payoffs, which increases the probability of securing better profits. To ensure that all the participating food trucks have high chances of acquiring better profits, standard deviation of the individual payoffs is also considered to select the best performing algorithm.

To test the quality of the solutions offered by the algorithms, the algorithms are run for different numbers of food trucks for a number of times, each time the initial positions of the food trucks are changed. However, the dataset containing the grid values was kept constant for all the tests. The experiments were performed on a 64-bit Arch-Linux desktop with an Intel Core2 Duo Processor E7500 having 2.93 GHz base frequency and 8 GB RAM. In order to get precise comparison of the time required, preprocessing of the datasets was performed before running each test. Table 1. shows the results obtained after running the algorithms for different test cases. The proposed algorithm clearly provides solutions that are more desirable i.e., with higher mean and lower standard deviation in all cases in comparison with the other algorithms. With the increase in the number of food trucks, a gradual decrease is observed in the mean payoffs of the solutions, but the proposed algorithm has higher mean in all the cases. The decrease in means with an increasing number of participating food trucks can be attributed to a larger search space. Higher numbers of participating food trucks can lead to a wider spread of

Table 1 Mean and standard deviation for the solutions obtained by the algorithms under experimentation over different numbers of food trucks

Number of vehicles	Proposed algorithm		Tabu Search		Genetic Algorithm	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
12	8.97647	2.54151	8.55388	2.939506	8.659874	2.954127
11	9.998775	2.468167	8.31585	2.482852	8.745142	2.854123
10	9.994358	2.116615	8.455711	2.562389	8.956821	2.741236
9	11.380883	1.584377	8.752024	2.533251	9.145721	2.523516
8	12.308324	0.927346	8.990925	2.589569	9.325687	2.496523
7	12.508012	0.881782	9.321708	2.605477	9.365874	2.425168
6	12.360655	0.868983	8.6433	2.167549	9.412587	2.396524
5	13.145097	0.464602	8.666387	2.673757	9.452163	2.365542
4	13.45098	0.244115	8.61376	2.651331	9.521489	2.225871
3	13.470588	0.279139	8.402323	1.523318	9.632598	2.103068

them over the region under study, which can bring more variation to the maximum score a food truck can get. It can be seen that the proposed heuristic produced solutions with considerably higher means until the number of food trucks reached 12, where it becomes very close to the means produced by the other two algorithms. This is followed by Genetic Algorithm.

A common trend that could be seen from the results is that for all the algorithms, the standard deviation kept on increasing with the increase in food trucks, which is reasonable because more participants in the system introduces more variability in the system, as they are starting off from different areas. Standard deviations of the solutions provided by Genetic Algorithm and Tabu Search nearly remained constant, even with the increase in the number of participating food trucks. In case of the proposed algorithm, a steady increase in the standard deviation values is observed, but they are still lower than the ones acquired from the other two methods. These results indicate that the proposed algorithm would be a better choice to obtain optimal results. As the proposed problem is a combinatorial optimization problem, its complexity is exponential [18]. Their complexity can reach exponential limits to find the optimum solution, rendering them extremely difficult to solve. Therefore, approximate solutions are used, but even these can take exponential time to be solved. The service range of the food truck also affects the time complexity of the algorithm, as a larger radius for the service area leads to larger search space, which can slow down the search. As can be seen from Fig. 2, despite processing the same dataset, the proposed algorithm produces results comparatively faster than genetic algorithm and tabu search. The solution search time is dependent on the area of study, the number of food trucks and their operating range. The search time increases exponentially with these parameters. The experiment was conducted only for a maximum number of 12 food trucks, as it was not possible to further evaluate the results because of slow performance. Genetic algorithm and tabu search showed comparable performance. Major optimizations and algorithm design changes have to be made to make this applicable for real-time scenarios. From these results, it can be concluded that the algorithm proposed in this paper is able to allocate spatio-temporal retail locations to food trucks in a cooperative-competitive arrangement which benefits all the participating food trucks, while maximizing the objective function for each individual truck, resulting in an allocated location which has a better consumer demand, less competition and is closer to the initial position of the food truck.

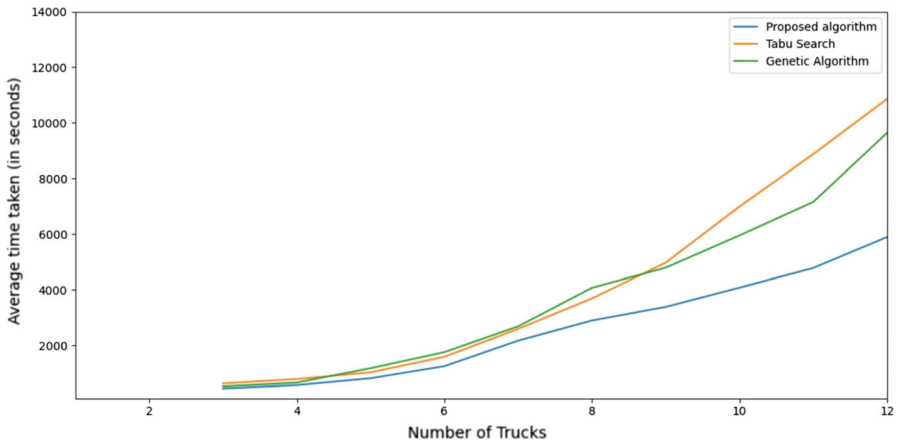


Fig. 2 Mean amount of time required by the algorithms

5 Conclusion and future works

In this paper, an approach has been introduced to recommend profitable spatio-temporal locations to food trucks with the aim of acquiring better profits and saving resources. An economic model is formulated, which uses various socio-economic parameters, to better identify the problem and derive meaningful results. Methods are discussed to find the profitability of an area using various time varying, location attributes. A game theoretic cooperative-competitive model is developed to model optimize the global solutions to the problem under discussion. The proposed algorithm is then compared with other optimisation techniques on the same datasets. Key evaluation metrics like mean and standard deviation of the agents' payoffs, and time taken to generate solutions were measured to show that the proposed methodology is better suited for profitability of the participating food trucks.

A huge scope is left in this field, concerning majorly the design of time efficient optimisation techniques. A real time internet based application can be developed to link the food trucks and validate the approach. Researchers can establish different economic models incorporating other parameters to get better predictions. Crowd movements, market history and trends can be tracked and integrated into the systems for improved labeling of profitable regions in a city. Introduction of geo-spatial analysis techniques can be used to derive real-time insights for ranking the locations for better profitability.

Data availability statement The data that support the findings of this study are available from [9] but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of [9].

Declaration

Conflict of interest We declare that there is no conflict of interest for this research submission.

References

1. 17 NEXUS 35 (2011-2012) The New Food Truck.
2. A comparative analysis of a Tabu Search and a Genetic Algorithm for solving a University Course Timetabling Problem. <https://www.diva-portal.org/smash/get/diva2:810264/FULLTEXT01.pdf> (for comparison)
3. A Cooperative-Competitive Multi-Agent Framework for Auto-bidding in Online Advertising. <https://doi.org/10.48550/arXiv.2106.06224>
4. Advocacy: Social Media, Mobile Food Vending Associations, Truck Lots, & Litigation in California & Beyond
5. Akbar T (2017) Research paper- economics and game theory Paper
6. Alfiero S, Giudice A, Bonadonna A (2017) Street food and innovation: the food truck phenomenon. *Br Food J* 119:00–00. <https://doi.org/10.1108/BFJ-03-2017-0179>
7. Andreica A, Chira C (2015) Best-order crossover for permutation-based evolutionary algorithms. *Appl Intell* 42:751–776. <https://doi.org/10.1007/s10489-014-0623-0>
8. Anenberg E, Kung E (2015) Information technology and product variety in the city: The case of food trucks. *J Urban Econ* 90:60–78. <https://doi.org/10.1016/j.jue.2015.09.006>
9. Application form (2021) Retrieved from <https://www.pmc.gov.in/en/census>
10. Bhandawat R Location Optimization for a Mobile Food Vendor Using Public Data
11. Burguillo J (2018) Game theory. https://doi.org/10.1007/978-3-319-69898-4_7
12. Caliori D (2020) Note on deterministic and stochastic choice regularity and property α
13. Cao K, Zhang W, Wang T (2019) Spatio-temporal land use multi-objective optimization: a case study in Central China. *Trans GIS* 23:726–744. <https://doi.org/10.1111/tgis.12535>
14. Chang X, Li J (2019) Business performance prediction in location-based social commerce. *Expert Syst Appl* 126:112–123. <https://doi.org/10.1016/j.eswa.2019.01.086>
15. Cho S, Lee G (2022) Spatiotemporal multi-objective optimization for competitive Mobile vendors' location and routing using De facto population demands. *Geogr Anal* 54:294–308. <https://doi.org/10.1111/gean.12292>
16. Choi H, Choi E-K, Yoon B, Joung H-W (2020) Understanding food truck customers: Selection attributes and customer segmentation. *Int J Hospit Manag* 90:102647. <https://doi.org/10.1016/j.ijhm.2020.102647>
17. Chua B-L, Ab Karim M, Lee S (2020) Customer Restaurant Choice: An Empirical Analysis of Restaurant Types and Eating-Out Occasions. *Int J Environ Res Public Health* 17:6276. <https://doi.org/10.3390/ijerph17176276>
18. Dammeyer F, Voß S (1993) Dynamic tabu list management using the reverse elimination method. *Ann Oper Res* 41:29–46. <https://doi.org/10.1007/BF02022561>
19. Gao X, Du H, Han M (2017) Combinatorial optimization and applications 11th international conference, COCOA 2017, Shanghai, China, December 16–18, 2017, proceedings. Springer International Publishing, Part I
20. Ghazikhani A, Mashhadi H, Monsefi R (2010) A novel algorithm for coalition formation in Multi-Agent Systems using cooperative game theory. Proceedings - 2010 18th Iranian conference on electrical engineering, ICEE 2010. <https://doi.org/10.1109/IRANIANCEE.2010.5507017>
21. Giocoli N (2004) Nash equilibrium. *History Political Econ* 36(4):639–666. <https://doi.org/10.1215/00182702-36-4-639>
22. Habet D (2009) Tabu search to solve real-life combinatorial optimization problems: a case of study. In: Abraham A, Hassanien AE, Siarry P, Engelbrecht A (eds) Foundations of computational intelligence volume 3 Studies in Computational
23. Hadzic S, Rodriguez J, Mumtaz S (2013) Cooperative game theory and its application in localization algorithms. INTECH Open Access Publisher.
24. Han SH, Nguyen B, Lee TJ (2015) Consumer-based chain restaurant brand equity, brand reputation, and brand trust. *Int J Hospit Manage* 50:84–93. <https://doi.org/10.1016/j.ijhm.2015.06.010>
25. Hegde S, Satyappanavar S, Setty S (2017) "Restaurant setup business analysis using yelp dataset," 2017 International Conference on Advances in Computing. Commun Inform (ICACCI):2342–2348. <https://doi.org/10.1109/ICACCI.2017.8126196>
26. Hines G, Rahwan T, Jennings N (2012) An anytime algorithm for finding the ϵ -Core in nontransferable utility coalitional games. ECAI
27. Huang S, Hu B, Liao R, Xiao J, He D, Guan Z (2019) "Multi-Agent Cooperative-Competitive Environment with Reinforcement Learning" .2019 IEEE 8th Data Driven Control Learn Syst Conf (DDCLS). 1382–1386. <https://doi.org/10.1109/DDCLS.2019.8909048>
28. Liu Q, Liu J, Liu D (2018) Intelligent multi-objective public Charging Station location with sustainable objectives. *Sustainability* 10(10):3760. <https://doi.org/10.3390/su10103760>

29. (2020) Maps of Pune, Maharashtra. Google Maps maps.google.com
30. Martin N (2014) Differential valuation of immigrant and 'gourmet' vendors in Chicago. *Int J Urban Reg Res* 38:1867–1883. <https://doi.org/10.1111/1468-2427.12169>
31. Maschler M, Solan E, Zamir S Coalitional games with transferable utility. *Game Theory*:659–685. <https://doi.org/10.1017/cbo9780511794216.017>
32. McCall J (2005) Genetic algorithms for modeling and optimisation. *J Comput Appl Math* 184:205–222. <https://doi.org/10.1016/j.cam.2004.07.034>
33. McNeil P, Young CA (2019) Customer satisfaction in gourmet food trucks: exploring attributes and their relationship with customer satisfaction. *J Foodserv Bus Res* 22(4):326–350. <https://doi.org/10.1080/15378020.2019.1614400>
34. Meng R, Ye Y, Xie N-g (2010) Multi-objective optimization design methods based on game theory. *Proceed World Congress Intell Control Autom (WCICA)*. <https://doi.org/10.1109/WCICA.2010.5554307>
35. Moon. (2011). Service demand of consumer with deterministic perceptions. https://doi.org/10.1007/978-3-642-20189-9_2.
36. Multi-objective game theory optimization for balancing economic, social and ecological benefits in the Three Gorges Reservoir operation
37. Norozpour S, Safaei M (2020) *IOP Conf. Ser.: Mater. Sci. Eng.* 993:012114
38. Nowe A, Vrancx P, Hauwere D, Yann-Michaël. (eds) (2012) In: *Game theory and multi-agent reinforcement learning*. https://doi.org/10.1007/978-3-642-27645-3_14
39. QGIS Development Team 2009. QGIS geographic information system. Open Source Geospatial Foundation. URL <http://qgis.org>
40. Radulescu R, Mannion P, Roijers DM, Nowé A (2020) Multi-objective multi-agent decision making: a utility-based analysis and survey. In *proceedings of the 19th international conference on autonomous agents and MultiAgent systems*. International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 2158–2160.
41. Sarkar S, Curado Malta M, Dutta A (2022) A survey on applications of coalition formation in multi-agent systems. *Concurrency Computat Pract Exper* 34(11):e6876. <https://doi.org/10.1002/cpe.6876>
42. Shams F, Luise M (2013) Basics of coalitional games with applications to communications and networking. *J Wireless Com Network* 2013:201. <https://doi.org/10.1186/1687-1499-2013-201>
43. Shin YH, Kim H, Severt K (2019) Consumer values and service quality perceptions of food truck experiences. *Int J Hospit Manag* 79:11–20. <https://doi.org/10.1016/j.ijhm.2018.12.008>
44. Sohrabi MK, Azgomi H (2020) A survey on the combined use of optimization methods and game theory. *Arch Computat Methods Eng* 27:59–80. <https://doi.org/10.1007/s11831-018-9300-5>
45. Trunfio GA (2006) Exploiting Spatio-temporal Data for the Multiobjective Optimization of Cellular Automata Models. In: Corchado, E., Yin, H., Botti, V., Fyfe, C. (eds) *Intelligent Data Engineering and Automated Learning – IDEAL 2006*. IDEAL 2006. Lecture notes in computer science, vol 4224. Springer, Berlin, Heidelberg doi: https://doi.org/10.1007/11875581_10
46. Uchigaito H et al (2021) Multi-objective Spatiotemporal Optimization of Transportation and Power Management by using Multiple Electric Vehicles in Nanogrid Networks. *2021 IEEE Electrical Power Ener Conf (EPEC)*:1–8. <https://doi.org/10.1109/EPEC52095.2021.9621698>
47. Wai B, Liu A, Huen L Selecting Optimal Locations of Food Trucks: A Case Study for the City of Vancouver
48. Wang F, Chen L, Pan W (2016) Where to place your next restaurant? Optimal Restaurant Placement via Leveraging User-Generated Rev. <https://doi.org/10.1145/2983323.2983696>
49. Wooders MH (1983) The epsilon core of a large replica game. *J Math Econ* 11(3):277–300. [https://doi.org/10.1016/0304-4068\(83\)90005-8](https://doi.org/10.1016/0304-4068(83)90005-8)

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.